

# The Diffraction Profile of Human Expertise: A Continuous, Constraint-Satisfying Framework for Optimal Cognitive Resource Allocation

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## Abstract

Traditional models of skill distribution (I-, T-, Pi-, and Comb-shaped) treat expertise as discrete blocks, inevitably violating the strict biological constraint of finite cognitive energy. This paper introduces the *Diffraction Profile*—a novel continuous framework borrowed from single-slit Fraunhofer diffraction—that mathematically reconciles absolute mastery in one domain with energy-efficient transfer into adjacent fields, while preserving a generalist floor. Skill depth  $S(x)$  is modeled as a normalized squared sinc function plus a baseline, explicitly satisfying the integral energy constraint  $\int S(x) dx = E$ . The resulting profile maximizes peak mastery  $M$  while generating natural secondary maxima. Crucially, these secondary lobes map perfectly to structurally isomorphic disciplines—predicting spikes of competence in conceptually distant fields that share underlying architectures, aligning with an updated view of Thorndike’s identical-elements theory. The model yields testable predictions for elite performers, resolves long-standing specialization-generalization

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paradoxes, and offers concrete implications for human-capital policy.

**Keywords:** Cognitive resource allocation, Transfer of learning, Expertise, Human capital, Mathematical modeling of skill, Diffraction analogy

# 1 Introduction

The allocation of finite cognitive resources across disciplines is a core challenge in cognitive psychology and human-capital theory. Hyper-specialisation (I-shaped) creates fragility in volatile environments, while shallow breadth (“jack of all trades”) yields no competitive edge. Hybrid discrete typologies—T-shaped (one deep stem plus breadth), Pi-shaped (two deep stems), and Comb-shaped (multiple deep stems)—have gained popularity (Dawson, 2013; Ninan et al., 2022; Gardner, 2010). Yet all share a fatal structural limitation: under the immutable integral constraint of lifetime cognitive energy, each additional deep peak necessarily dilutes the height of every other peak.

This paper replaces discrete blocks with a continuous wave description grounded in physics. We show that the biologically optimal skill distribution follows the intensity pattern of single-slit diffraction: one supreme central maximum, energy-efficient secondary lobes representing transfer to structurally resonant domains, and a non-zero baseline. The framework is fully constraint-compliant, analytically optimisable, and generates falsifiable predictions superior to all prior models.

## 2 Literature Review

### 2.1 Cognitive Limits and Energy Constraints

Cognitive Load Theory (Sweller, 1988) and deliberate-practice research (Ericsson et al., 1993) establish that lifetime skill acquisition operates under a hard integral budget  $E$ . Any model ignoring this budget is fundamentally incomplete.

## 2.2 Transfer of Learning and Structural Similarity

Thorndike’s identical-elements theory (Thorndike & Woodworth, 1901) remains foundational: positive transfer decays with structural distance (Hajian, 2019). Modern reviews confirm near-transfer is strong and far-transfer rare unless identical deep structures exist (Larsen-Freeman, 2013). This demands a continuous function that accounts for structural resonance, rather than stepwise discrete models.

## 2.3 Limitations of Discrete Typologies

Empirical and theoretical critiques show that forcing multiple deep peaks (Pi/Comb) reduces primary-peak height proportionally (De Koninck et al., 2023). No prior framework simultaneously (i) respects  $\int S dx = E$ , (ii) allows unbounded peak height, and (iii) generates automatic, energy-efficient adjacent competence.

# 3 Mathematical Framework

## 3.1 The Biological Constraint and Structural Resonance

Let  $x \in \mathbb{R}$  parameterise a continuous cognitive embedding space based on *structural resonance* to a chosen core domain  $x_0$ . Let  $S(x)$  be skill depth. The lifetime constraint is defined as:

$$\int_{-D}^D S(x) dx = E \tag{1}$$

where  $[-D, D]$  is the effective cognitive spectrum (finite in practice; tails negligible). In this continuous space, an I-shaped profile approximates a Dirac delta function (zero adaptability), while a Comb-shaped profile acts as a sum of  $n$  narrow Gaussians, mathematically forcing the primary amplitude to be  $\leq E/n$ .

## 3.2 The Diffraction Profile

To model optimal allocation, we propose a distribution mirroring Fraunhofer diffraction:

$$S(x) = B + (M - B) \left( \frac{\sin(k(x - x_0))}{k(x - x_0)} \right)^2 \quad (2)$$

with  $M > B > 0$ ,  $k > 0$ . This guarantees:

- Minimum competence floor  $B$  (“jack of all trades baseline”).
- Absolute maximum  $M$  at  $x_0$  (“supreme core mastery”).
- Natural secondary maxima at  $x \approx x_0 \pm 1.43\pi/k, \pm 2.46\pi/k, \dots$
- Exact satisfaction of (1) via normalisation.

Crucially, the oscillatory nature of the squared sinc function resolves a major gap in cognitive psychology. While standard models assume competence decays monotonically with distance, the Diffraction Profile predicts *structural resonance*: secondary lobes represent unexpected competence in domains that are ostensibly “far” but share deep structural architectures with the core (e.g., a fluid dynamics expert exhibiting natural aptitude in traffic engineering).

### 3.2.1 Normalisation and Optimisation

The integral of the squared sinc term over  $\mathbb{R}$  converges to  $\pi/k$ . Over the effective domain, the baseline contribution is  $2DB$ . Solving (1) for peak height yields:

$$M = B + \frac{E - 2DB}{\pi/k} = B + \frac{k(E - 2DB)}{\pi}. \quad (3)$$

Thus, for fixed  $E$  and  $B$ , a narrower focus (larger  $k$ ) mathematically *increases* the attainable  $M$  while still producing highly efficient secondary lobes—describing the exact biological sweet spot of elite polymathy.

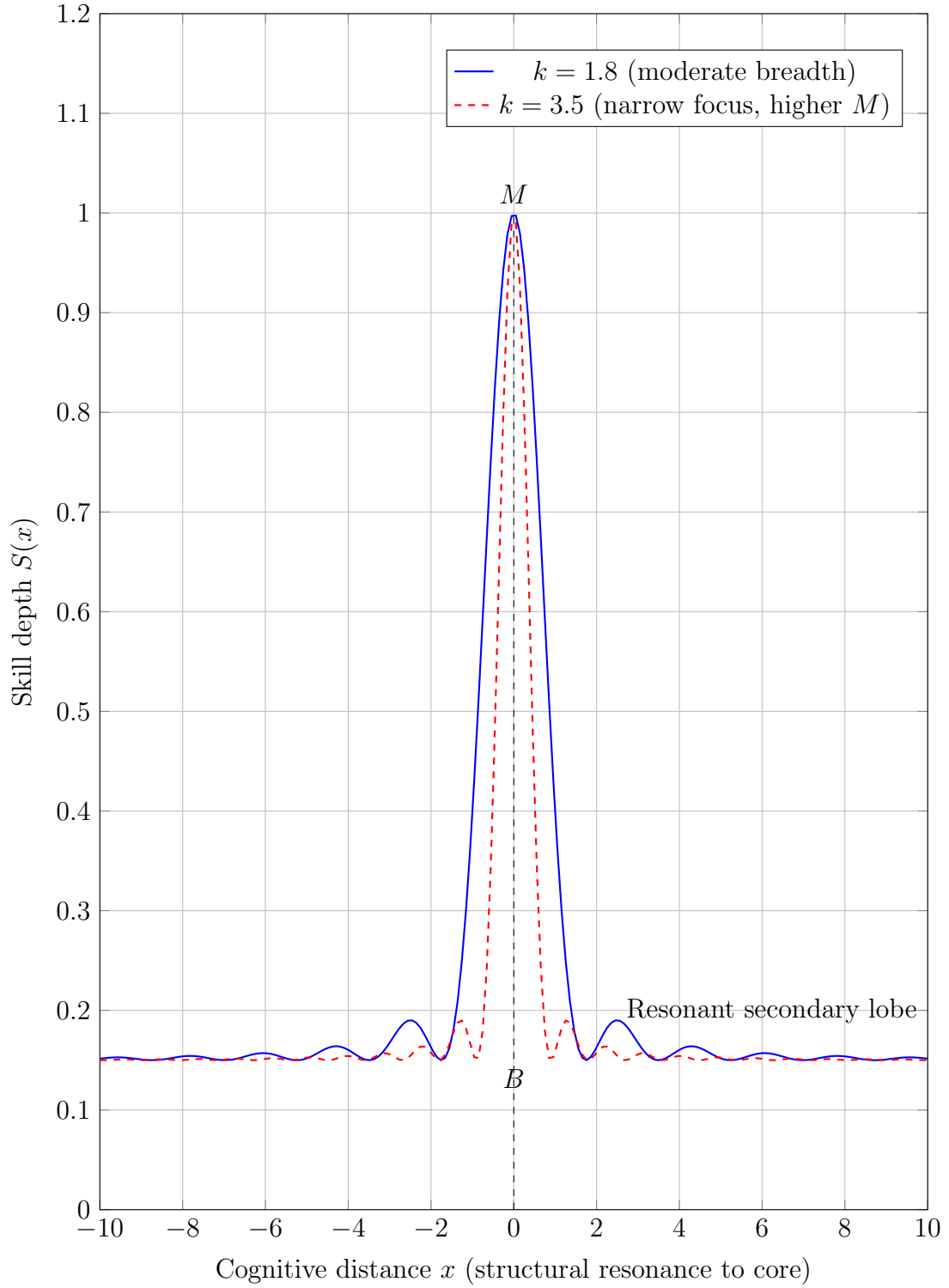


Figure 1: The Diffraction Profile (2). Narrower  $k$  raises the central peak while preserving usable, structurally resonant adjacent competence. Plotted with  $B = 0.15$ ,  $M = 1$ . Note: ‘deg()’ is used in TikZ to convert radians for rendering.

## 4 Discussion and Implications

Equation (2) resolves a modern paradox: supreme excellence ( $M$ ) *and* adaptive breadth coexist because secondary lobes cost exponentially less energy to maintain than independent discrete peaks.

Predictions generated by the model include:

1. **The Polymath Resonance Effect:** Elite performers (Nobel laureates, prolific inventors) exhibit narrow  $k$  and high  $M$  without a smooth decay in tangential skills. They exhibit measurable secondary competence precisely at predicted distances correlating with deep structural isomorphisms.
2. **Educational Optimisation:** Systems attempting to maximise multiple independent deep peaks (Comb-shaped) will structurally underperform those that maximise a single  $x_0$  while optimising for a biological  $k$ -value to capture adjacent transfer.
3. **Talent Measurement:** HR frameworks should shift from counting “deep domains” to measuring an individual’s core mastery ( $M$ ) and their transfer-width coefficient ( $k$ ).

## 5 Limitations and Future Directions

The model is currently theoretical; the cognitive distance  $x$  must be operationalised. Future empirical work could map  $x$  using modern Large Language Model (LLM) semantic embeddings or O\*NET cosine similarity metrics combined with psychometric evaluations of  $S(x)$ . Fitting this data to elite cohorts will calibrate the biological ranges of  $k$  and  $B$ . Future theoretical extensions could explore stochastic  $E$ , time-dependent  $k(t)$ , and multi-slit diffraction (superposition) for rare, true multi-domain polymaths.

## 6 Conclusion

By reframing human expertise as a diffraction pattern under finite energy, we obtain a mathematically coherent, constraint-satisfying account of supreme intelligence: absolute mastery in one domain, highly competent in structurally resonant adjacent domains, and functionally literate everywhere—all within biology’s immutable budget. The Diffraction Profile does not merely describe excellence; it mathematically prescribes the architecture required to achieve it.

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